

A Test of Mean Resolution Time

Unit 4 · Topic 4.5 · TODAY'S PROBLEM

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

Leah: “The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide.”

Damon: “Because $p > 0.05$, accept H_0 . The test proves the population mean was exactly 12 hours and the observed difference was only chance.”

Ticket sample

Summary	Value
Population	1200
n	24
Mean (h)	11.1
SD (h)	2.8
Shape	Symmetric; no outliers

FIRST TAKE (5 min, on your sheet)

Does this small study seem able to prove that the average was exactly 12 hours? Explain in one sentence.

- DOK 1** (a) State H_0 and H_a for the population mean resolution time.
- DOK 2** (b) Using $\bar{x} = 11.1$, $s = 2.8$, and $n = 24$, calculate t . Use the supplied $p = 0.0645$; you do not need to find it again.
- DOK 3** (c) In 3–4 sentences, decide whose account is justified. Compare the p-value with 0.05, state the conclusion for all 1,200 tickets, and explain why failure to reject does not prove equality or that chance was the only explanation.

Video + follow-along: [u7_lesson5_live.html](https://www.khanacademy.org/a/u7_lesson5_live_html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

